Piyush Makhija

+91 9406888655 | piyushmakhija26@gmail.com | linkedin.com/in/piyush-makhija
github.com/PiyushMakhija26 | leetcode.com/u/piyushmakhija | Portfolio

Summary

Computer Science undergraduate specializing in backend engineering, distributed systems, and AI/ML. Hands-on experience building high-concurrency gateways, consensus algorithms (Raft), real-time collaborative applications, and scalable infrastructure using Go, Python, and JavaScript.

Education

VIT Bhopal University — Bhopal, MP
Bachelor of Technology in Computer Science (AI & Machine Learning) CGPA : 8.83 — *2023 – 2027*

Technical Skills

Languages:C++ ,C ,Python ,JavaScript ,SQL ,Go
Frontend, Backend & Databases: React, FastAPI, Node.js, PostgreSQL, Redis
AI & Machine Learning: RAG, Prompt Engineering
Infrastructure & Tools: Git, Docker, Prometheus, Grafana, GitHub Actions, WebSockets, REST APIs
Computer Science Concepts: Distributed Systems (Raft, WAL), Concurrency, Operational Transformation, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS

Featured Projects

Collaborative Document Editor | *React, Node.js, WebSockets, PostgreSQL, Redis* — GitHub
- Implemented Operational Transformation with property-based testing, ensuring 100% convergence across 1,000 concurrent edits.
- Built a Jupiter-style sync protocol achieving 6.4ms avg latency for 100 concurrent editors with a 99.98% success rate.
- Scaled horizontally via Redis Pub/Sub sharding with 8.19ms overhead, ensuring zero data loss during offline reconnections.

API Gateway | *FastAPI, Redis, Docker, Prometheus, GitHub Actions* — GitHub
- Architected a FastAPI reverse proxy gateway sustaining 300+ req/sec at sub-600ms p99 latency with CI/CD and monitoring.
- Developed an atomic Redis rate limiter via Lua (shedding 89% burst traffic) and a circuit breaker with <150ms failover.

OmniChat | *Python, FastAPI, PostgreSQL, Redis, Docker* — GitHub
- Engineered a multi-LLM routing engine with auto-failovers, cutting API costs by 96.7% and ensuring 100% uptime at 1.05s latency.
- Built a hybrid RAG pipeline backed by a dual-layer memory system for context retention and secured via refresh tokens.

Distributed Key-Value Store | *Go, Raft Consensus, WAL* — GitHub
- Built Raft consensus in Go (leader election, replication) across a 3-node cluster, achieving 288ms failovers and 184 writes/sec.
- Enforced quorum consistency to prevent split-brain and implemented CRC32-checksummed WAL for rapid crash recovery.

Focus Lock | *JavaScript, Chrome Extension API, IndexedDB* — Live | GitHub
- Published a privacy-first Chrome extension featuring distraction blocking, YouTube filtering, and IndexedDB local analytics.

Certifications

Applied Machine Learning (Michigan) | Google IT Support | Google Data Analytics | Microsoft SC-900

Achievements

ET-AI Hackathon 2026: Selected as a Semi-Finalist among participating teams.
LeetCode: Solved 100+ problems, demonstrating consistent problem-solving and DSA practice.